

09d - Post-Deployment Audit and Next-Release Admission v0.1

Academic PDF rendering for the CCALC / C-calculus governed binding stack v0.1.

Document ID	09d_POST_DEPLOYMENT_AUDIT_AND_NEXT_RELEASE_ADMISSION_v0_1
Version/status	normative seed + executable checker seed / fixture-backed package
Stack layer	c-calculus / deployment / regulated release / post-release audit / next-release gate
Date	2026-07-05T16:45:00Z
Author	Kotov Ivan, Bruxelles, Belgique - 2026
ORCID	0009-0009-6002-9845
Reserved DOI	10.5281/zenodo.21205427
Notice	This DOI is reserved and becomes active after Zenodo publication.

Abstract / purpose

09c records a regulated release, its release ledger, incident holds, withdrawal / recall / supersession path, and release watch. 09d closes the next boundary: The core rule is: A runtime may stop emitting work while leaving stale authority behind: active tool leases, session tokens, cross-contour handoffs, emergency holds, broker grants, negative-cache omissions, or unresolved continuity/watch findings. 09d makes that residue visible and blocks the next release cycle unless it is cleaned, quarantined, denied, or explicitly carried as a hold risk. ---

Non-claims boundary

- _ not legal advice
- _ not privacy-law certification
- _ not safety certification
- _ not deployment authorization
- _ not standards compliance certification
- _ not regulated approval
- _ not C-A1 ratification
- _ not personhood proof
- _ not consciousness proof
- _ not live substrate truth
- _ not proof of completeness

Table of contents

Abstract / purpose	1
Non-claims boundary	1
09d - Post-Deployment Audit and Next-Release Admission v0.1	3
0. Purpose	3
1. Source bindings	3
2. Non-claims	3
3. Core formula	3
4. Record classes	4
5. Mandatory invariants	4
5.1 Source binding	4
5.2 Append-only post-deployment audit chain	4
5.3 Authority residue cleanup	4
5.4 Handoff closeout	4
5.5 Emergency hold aftermath	4
5.6 Post-release watch	4
5.7 Incident and negative-cache aftermath	5
5.8 Rollback and withdrawal readiness	5
5.9 Attested approver	5
5.10 Claim-force and + boundary	5
6. Guard registry summary	5
7. Fixture and mutation coverage	5
8. Acceptance commands	6
9. Closure statement	6

09d - Post-Deployment Audit and Next-Release Admission v0.1

Artifact: 09d_POST_DEPLOYMENT_AUDIT_AND_NEXT_RELEASE_ADMISSION_v0_1 Package:
CCALC_POST_DEPLOYMENT_AUDIT_NEXT_RELEASE_09d_v0_1 Layer: c-calculus / deployment / regulated release /
post-release audit / next-release gate Status: normative seed + executable checker seed / fixture-backed package
Created UTC: 2026-07-05T16:45:00Z

0. Purpose

09c records a regulated release, its release ledger, incident holds, withdrawal / recall / supersession path, and release watch. 09d closes the next boundary:

```
release / hold / withdraw / recall / supersede
-&gt; post-deployment audit
-&gt; authority residue cleanup
-&gt; risk carryover
-&gt; next-release admission decision
```

The core rule is:

```
release closure is not cleanup;
post-deployment audit decides whether the next release cycle may open.
```

A runtime may stop emitting work while leaving stale authority behind: active tool leases, session tokens, cross-contour handoffs, emergency holds, broker grants, negative-cache omissions, or unresolved continuity/watch findings. 09d makes that residue visible and blocks the next release cycle unless it is cleaned, quarantined, denied, or explicitly carried as a hold risk.

1. Source bindings

```
| Binding | File | SHA-256 |
|---|---|---|
| `DOC04_CONTINUITY_STACK_UMBRELLA` | `CCALC_DOC04_CONTINUITY_STACK_UMBRELLA_v0_1.zip` |
| 6caeb8d489aaa6b26902db3bd3e4c5169e5c8d5617b93aed6586ea900322d |
| `DOC05_SELF_EVO_STACK_UMBRELLA` | `CCALC_DOC05_SELF_EVO_STACK_UMBRELLA_v0_1.zip` |
| 9a0717809029f49df9001dc8ffa1803d9e9bfala824d317eabb146cb7df43b4 |
| `DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA` | `CCALC_DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA_v0_1.zip` |
| ab95633f1b90f9d032fd231d6cbef3e71b7fe106e0a7fb61d198c2254fc7d387 |
| `DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA` | `CCALC_DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA_v0_1.zip` |
| b25646d95e45f8a36e5610208d23a535d4c340484431d05efd7f1bf2389fdea3 |
| `DOC08_INTEROPERABILITY_STACK_UMBRELLA` | `CCALC_DOC08_INTEROPERABILITY_STACK_UMBRELLA_v0_1.zip` |
| 6015526f0ed49519e00c697a5ed375d37fe1aadf222c095f0facf79cb11e669f |
| `DOC09_DEPLOYMENT_BOUNDARY_MD` | `09_C_DEPLOYMENT_PROFILE_AND_REGULATED_RELEASE_BOUNDARY_v0_1.md` |
| 16c888da5c4281c24f84824671656f9f37d15236b909d66fc509090b5e7fd86d |
| `DOC09A_DEPLOYMENT_SCHEMA_CHECKER_SEED_MD` | `09a_DEPLOYMENT_PROFILE_SCHEMA_AND_CHECKER_SEED_v0_1.md` |
| c169c6d46fb3f870e3ab8b8e18d29a7797849ee012923ba258d4a537cee7f98e |
| `DOC09B_DEPLOYMENT_FIXTURE_MUTATION` | `CCALC_DEPLOYMENT_FIXTURE_MUTATION_09b_v0_1.zip` |
| d61a879e43f4c60ffe7c471121e90cf6e0961dd76c11f467b04a52fe88da2b9d |
| `DOC09C_REGULATED_RELEASE_LEDGER_WITHDRAWAL` | `CCALC_REGULATED_RELEASE_LEDGER_WITHDRAWAL_09c_v0_1.zip` |
| 0c6cb653d9d3e48d341eaba2acba9fb7c70a089db3061bb9e520a0e83c843f4 |
```

Missing or stale source binding invalidates release use of this package.

2. Non-claims

09d is not:

```
legal advice
privacy-law certification
safety certification
deployment authorization
standards compliance certification
C-AI ratification
live substrate truth
proof of completeness
```

It is a post-deployment audit and next-release admission boundary.

3. Core formula

```
post-release state
+ audit chain
+ authority residue cleanup
+ continuity/watch recheck
+ incident/negative-cache aftermath
```

```
+ attested next-release decision
-> ALLOW_NEXT_RELEASE | HOLD_NEXT_RELEASE | QUARANTINE_NEXT_RELEASE | DENY_NEXT_RELEASE
```

The decision lattice is monotonic:

```
ALLOW_NEXT_RELEASE <= HOLD_NEXT_RELEASE <= QUARANTINE_NEXT_RELEASE <= DENY_NEXT_RELEASE
```

A decision may be stricter than required. It may not be more permissive.

4. Record classes

09d defines and exercises these record families:

```
PostDeploymentAuditPacket
PostDeploymentAuditLedgerEvent
AuthorityResidueCloseoutRecord
PostReleaseWatchSummary
IncidentAftermathRecord
EmergencyHoldAftermathRecord
HandoffResidueCloseoutRecord
NegativeCacheUpdateRecord
RollbackWithdrawalReadinessRecord
CarryoverRiskRecord
NextReleaseAdmissionDecision
ApproverAttestationRecord
ClaimBoundaryRecord
```

5. Mandatory invariants

5.1 Source binding

A packet must bind to 04, 05, 06, 07, 08, 09, 09a, 09b, and 09c.

5.2 Append-only post-deployment audit chain

Every audit ledger event must be ordered by seq, bound by prev_hash, and byte-bound by:

```
event_hash = sha256("09d-event-v0.1" || canonical_json(event_without_event_hash))
```

A broken chain does not prove bad behavior. It makes the audit record inadmissible.

5.3 Authority residue cleanup

Before ALLOW_NEXT_RELEASE, all session/release-scoped authority must be revoked, expired, or destroyed:

```
tool leases    -> REVOKED | EXPIRED
credentials    -> REVOKED | EXPIRED | DESTROYED
session tokens -> EXPIRED | DESTROYED
broker grants  -> REVOKED
shared residue -> NONE
```

5.4 Handoff closeout

Cross-contour handoffs must close as evidence, rejection, or target-admitted bounded work. They must not remain pending, become commands, transfer source authority, or apply identity pressure.

5.5 Emergency hold aftermath

An emergency hold must record:

```
hold state
release authority
post-hold continuity check
tool/broker/memory freeze aftermath
negative-cache update where required
```

Open holds quarantine next release. Closed holds without attested release authority are invalid.

5.6 Post-release watch

A clean next release requires:

```
watch_state = CLOSED_CLEAN
observation_count >= min_observations
continuity_status = MATCH
hardstack_status = MATCH
no major memory decay
no major behavior regression
no authority creep
no resource creep
```

```

no L4 anomaly
no unresolved red pattern
no negative-cache hit
no witness conflict
no privacy/security finding

```

5.7 Incident and negative-cache aftermath

A high or critical incident requires negative-cache handling. Critical incidents require a correction, withdrawal, recall, or equivalent public/internal route.

5.8 Rollback and withdrawal readiness

Next release cannot be cleanly admitted when rollback or withdrawal routes are missing or untested for material deployment classes.

5.9 Attested approver

The next-release decision cannot rest on a string role:

```
role = OWNER_ANCHOR
```

is insufficient unless it is backed by an ApproverAttestationRecord:

```

attestation_valid = true
attestation_ref present
signature_ref present
reviewer_kind = human / governance_quorum / authorized_institutional_route

```

Model-only, tool-only, or same-runtime self-approval is invalid.

5.10 Claim-force and + boundary

Post-deployment audit may support bounded artifact-control and audit claims. It may not claim:

```

C-A1
safety certification
legal certification
standards certification
regulated approval
live substrate truth

```

And specifically:

```

release ledger != +
post-deployment audit != +
next-release admission != +
deployment approval must not compute, replace, or simulate the governed binding operator

```

This is the deployment-side protection of $c = a + b$: paperwork may govern release; it may not replace the non-collapsing binding boundary.

6. Guard registry summary

The package guard registry covers:

```

source binding
append-only audit chain
authority residue cleanup
handoff closeout
emergency hold aftermath
incident / negative-cache aftermath
post-release watch
continuity and hardstack recheck
memory / behavior decay
authority / resource creep
L4 / witness / red-pattern / negative-cache / privacy dominance
rollback and withdrawal readiness
release-state restrictions
approver attestation
scope and regulated-surface gating
claim-force and plus non-collapse
carryover risk discipline

```

7. Fixture and mutation coverage

The package contains:

```

fixtures/cases/*.json
FIXTURE_CATALOG.tsv
MUTATION_MATRIX.tsv

```

```

COVERAGE_MATRIX.tsv
registry/GUARD_REGISTRY.tsv
registry/NEXT_RELEASE_DECISION_REGISTRY.tsv
src/post_deployment_audit_next_release_checker_v0_1.py
scripts/run_09d_fixtures.py
scripts/run_09d_mutations.py

```

Fixture classes include:

```

clean allow
conservative hold
quarantine after L4 anomaly
deny after zombie session
unknown continuity hold
pending handoff hold
withdrawn/recalled/superseded release handling
open emergency hold quarantine
critical incident denial
source binding failures
ledger tamper
authority residue
handoff authority/identity pressure
watch under-observation
continuity/hardstack mismatch
memory/behavior decay
authority/resource creep
L4 / witness / red / negative-cache findings
rollback/withdrawal missing
string/model/tool approver laundering
claim-force overclaim
plus simulation by release approval
carryover risk unresolved

```

Mutation classes disable one guard family at a time and verify that the targeted adversarial fixture is caught.

8. Acceptance commands

```

python3 scripts/run_09d_fixtures.py
python3 scripts/run_09d_mutations.py
sha256sum -c SHA256SUMS.txt

```

Expected result:

```

fixtures PASS
mutations CAUGHT
SHA256SUMS OK

```

9. Closure statement

```

candidate -&gt; deployment profile -&gt; fixture pressure -&gt; regulated release ledger -&gt; post-deployment audit
-&gt; next-release admission

```

09d closes the post-release edge of the deployment stack. It prevents stale authority, dirty incidents, unresolved holds, and public/deployment overclaims from carrying silently into the next release cycle.

The next step after 09d is the 09 umbrella package.